Battery Management System — Component Connection Diagram

Full wiring reference · 2× 18650 2500mAh 3.7V (2S = 7.4V nominal / 8.4V full) with pin connections and precautions



2S 8.4V USB-C CHARGER

STEP 1

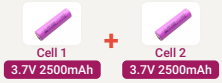
2S 8.4V USB-C Charger Module

OUT+ | OUT− | USB-C IN

- **OUT+** → P+ on BMS (HX-2S-A2)
- **OUT−** → P− on BMS (HX-2S-A2)

! Precaution: Never connect charger OUT to B+ / B−. Always verify polarity before plugging USB-C power.

OUT+ / OUT− → 8.4V charging



2S SERIES = 7.4V / 8.4V FULL

STEP 2 — Battery Pack → BMS Module

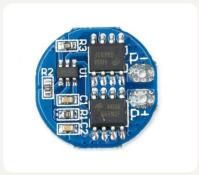
2× 18650 Li-ion 3.7V 2500mAh (2S) → HX-2S-A2 BMS Protection Plate

B+ | BM (mid) | B− | P+ | P−

- Cell 1 (+) = Pack+ → **B+** (BMS)
- Cell 1 (−) / Cell 2 (+) junction → **BM** midpoint (BMS)
- Cell 2 (−) = Pack− → **B−** (BMS)

! Precaution: Cells must be in **series (2S)**. Cell 1 (−) connects to Cell 2 (+) and also to BM. Wrong BM connection causes cell imbalance and damage. Total capacity: 2500mAh at 7.4V.

P+ / P− → Protected 8.4V out



HX-2S-A2 BMS PLATE

STEP 2B — BMS Details

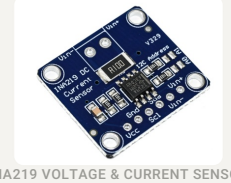
HX-2S-A2 Circular 2S 8.4V BMS 18650 Protection Plate

B+ | BM | B− | P+ | P−

- **P+** output → INA219 **VIN+**
- **P−** output → INA219 **VIN−**
- Charger **OUT+/OUT−** also connect to **P+/P−** for charging path

! Precaution: BMS handles over-charge, over-discharge, and short-circuit protection. Do not bypass B+/B− directly to load.

P+ / P− → 8.4V → INA219



INA219 VOLTAGE & CURRENT SENSOR

STEP 3 — BMS → INA219

INA219 — I²C Voltage & Current Sensor

VIN+ VIN− SCL SDA GND VCC

- P+ (BMS) → VIN+ (INA219)
- P− (BMS) → VIN− (INA219)
- MCU SCL / SDA → INA219 I²C pins for monitoring

! Precaution: Do NOT reverse VIN+ and VIN−. Keep wires short to reduce noise in current readings.

VIN+ / VIN− → 8.4V → buck input



LM2596S DC-DC BUCK

STEP 4 — INA219 → LM2596S Buck Converter

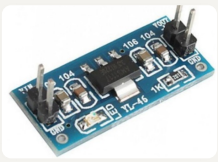
LM2596S DC-DC Step-Down Converter

IN+ IN− OUT+ OUT−

- VIN+ (INA219) → IN+ (LM2596S)
- VIN− (INA219) → IN− (LM2596S)
- Adjust trim pot → set OUT+ = ~6.5V before next stage

! Precaution: Before connecting AMS1117 — power ON and use a multimeter to adjust output to ~6.5V using the onboard trim potentiometer.

OUT+ / OUT− → ~6.5V regulated



AMS1117 5V REGULATOR

STEP 5 — LM2596S → AMS1117

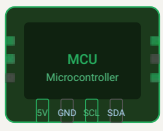
AMS1117 — 5V LDO Voltage Regulator

IN GND OUT (5V)

- OUT+ (LM2596S) → IN (AMS1117)
- OUT− (LM2596S) → GND (AMS1117)
- OUT (AMS1117) → MCU / System 5V rail

! Precaution: Input must be >6.2V (dropout margin). Do not exceed 500mA continuous current. AMS1117 gets warm — ensure ventilation.

5V stable output → 5V to MCU / load



MCU / SYSTEM LOAD

STEP 6 — Final Output

MCU / System Load

5V IN GND SCL (INA219) SDA (INA219)

- AMS1117 OUT → MCU 5V power pin
- AMS1117 GND → System GND (common ground)
- MCU SCL/SDA → INA219 for real-time power monitoring

Note: Tie all GNDs together (INA219, LM2596S, AMS1117, MCU) as a common ground reference for accurate measurements.

Cell Pack Specification — 2S Configuration

Cell type: 18650 Li-ion
Cells: 2× in Series (2S)
Cell voltage: 3.7V each (nominal)
Pack voltage: 7.4V nominal / 8.4V full
Capacity: 2500mAh (same as single cell)
Energy: ~18.5Wh total pack

Wire legend: Positive / V+ Negative / GND Regulated output I²C data (SCL/SDA) Battery midpoint (BM)